Developer Technology Trends: What Developers Use Today and Want Next

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Executive Summary

Purpose: Analysed Stack Overflow Developer Survey data to see which technologies developers use today and want next

Approach: Cleaned the data, explored it with SQL and Python, and built a Cognos Analytics dashboard

Key findings: JavaScript and SQL lead languages, PostgreSQL leads databases and AWS leads platforms

Future interest: Rust and Go enter the top 10 wanted languages and PostgreSQL grows as the most wanted database

Recommendation: Prioritise JavaScript/TypeScript, SQL, Python, PostgreSQL and AWS; watch Rust, Go, Supabase and FastAPI

14,943

JavaScript: most used language

11,514

PostgreSQL: most used database

12,193

PostgreSQL: most wanted database

Introduction

Background

Technology changes fast, so developers and organisations need to know which tools are in use and which are gaining interest

Data source

The Stack Overflow Developer Survey, with responses from developers around the world

Objective

Identify current and future technology trends and understand who the respondents are

Key questions

Which languages, databases, platforms and web frameworks are used today and wanted next?

Who are the respondents?

Methodology

1 Data

Notebooks use the full survey (65,437 responses); the dashboard uses a course extract of 18,845

2 Preparation

Cleaned with Python (Pandas): removed duplicates and handled missing values

3 Analysis

SQL and Python in Jupyter Notebooks to explore and count technology use

4 Visualisation

Cognos Analytics dashboard with three tabs: current use, future trends, demographics

Results

Current usage

JavaScript (14,943), SQL (12,602) and HTML/CSS (12,410) are the most used languages

PostgreSQL (11,514) leads databases; AWS (10,871) leads platforms

Future interest

JavaScript, SQL, TypeScript and HTML/CSS are the most wanted languages

PostgreSQL rises to 12,193; Rust and Go enter the top 10

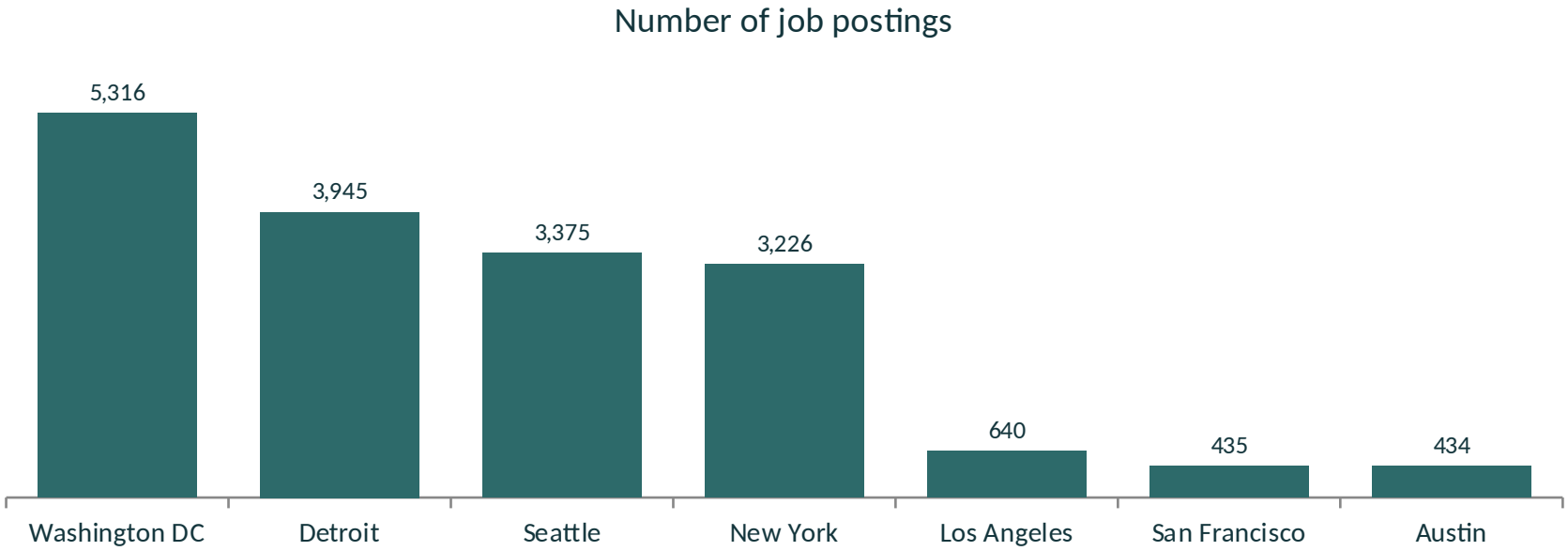
Demographics

41.3% are aged 25-34 and 27.3% are 35-44

Most common education: bachelor's degree (8,629)

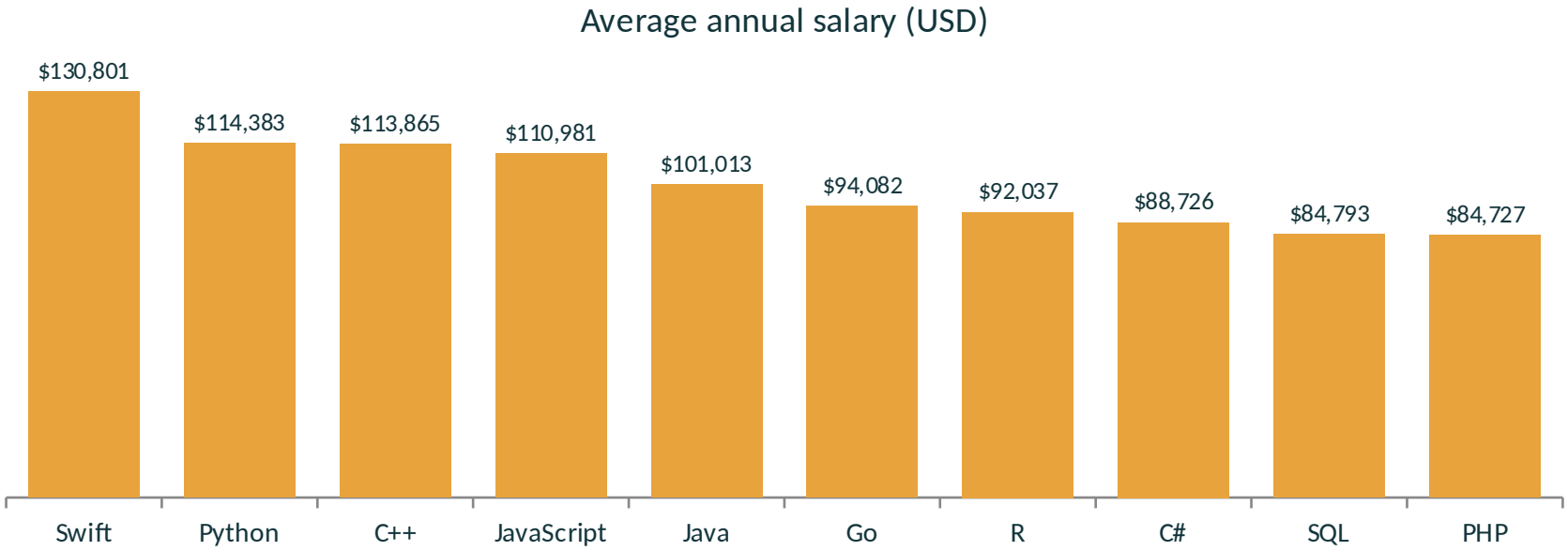
Most respondents: USA (3,441), Germany, India

Job Postings by Location



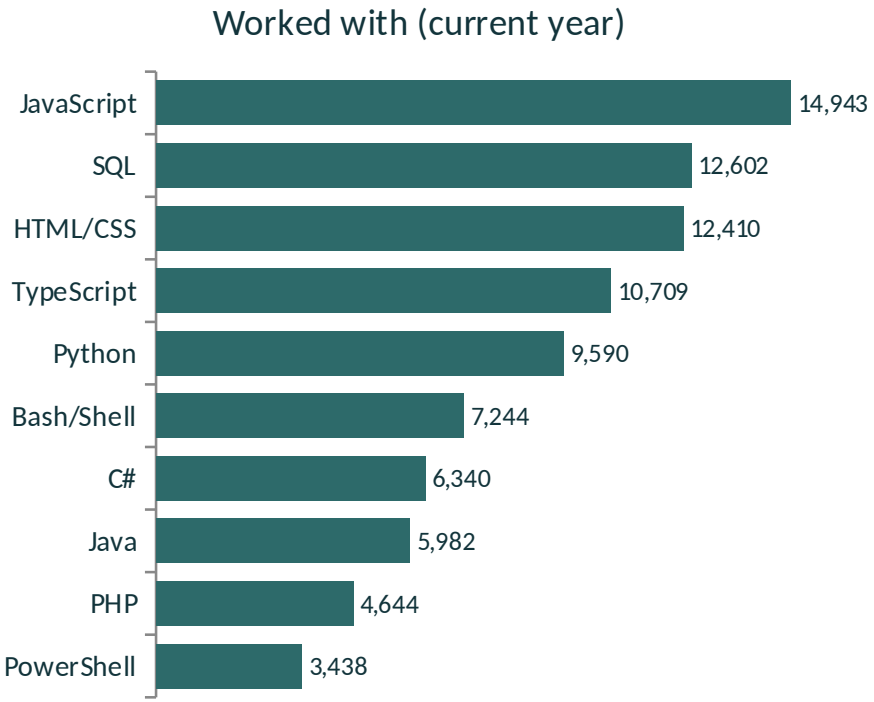
Washington DC has the most postings (5,316); Austin and San Francisco have the fewest. Source: jobs dataset provided with the course.

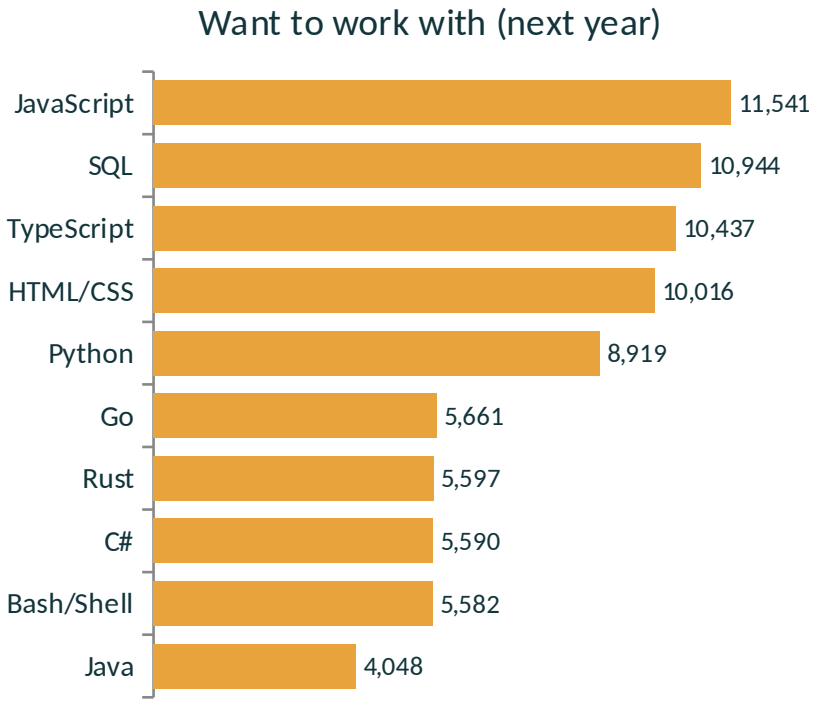
Average Annual Salary by Language



Swift pays the most ($130,801). Source: static web page provided with the course, scraped with BeautifulSoup, so figures are illustrative.

Programming Language Trends





Language Trends: Findings & Implications

Findings

JavaScript (14,943), SQL (12,602) and HTML/CSS (12,410) are the most used; JavaScript is also the most wanted (11,541)

TypeScript (10,709 to 10,437) and Python (9,590 to 8,919) keep most of their interest

Rust (2,284 to 5,597) and Go (3,305 to 5,661) rise; Java falls from 5,982 to 4,048

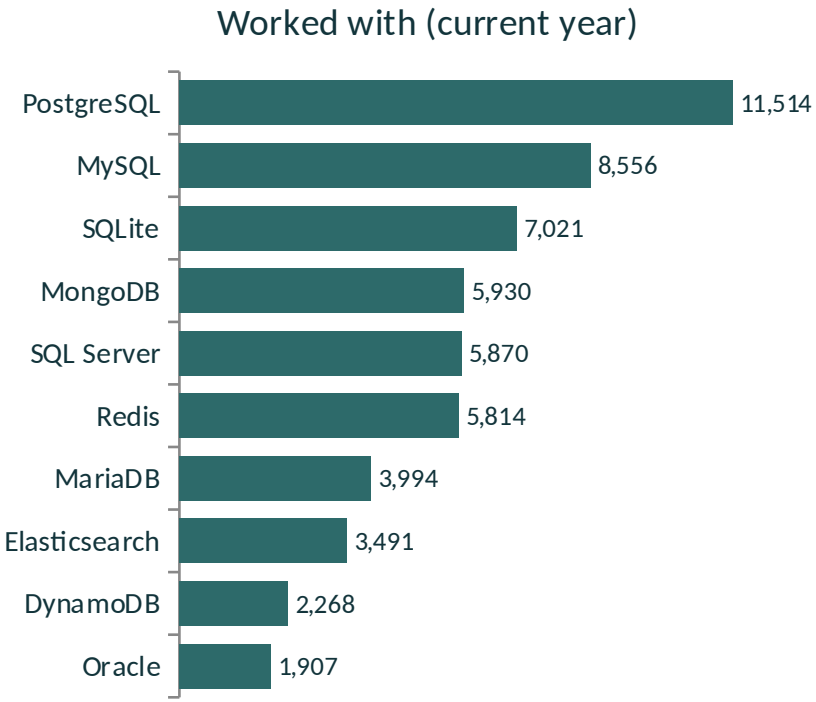
Implications

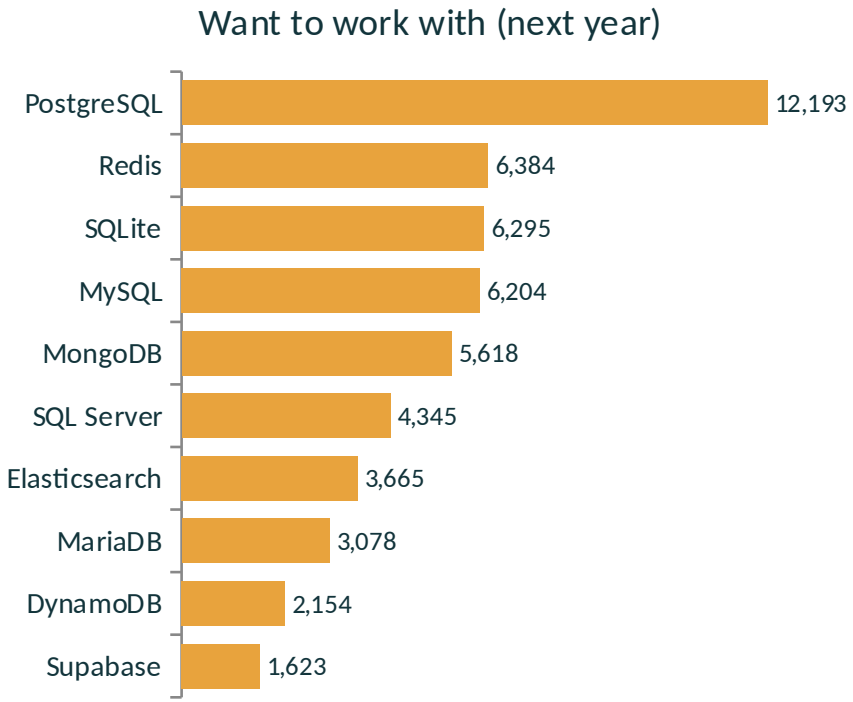
JavaScript, SQL and HTML/CSS remain the safest core skills to build on

TypeScript and Python are strong additions to that core

Rust and Go are worth watching, and lower interest in Java suggests it may be cooling

Database Trends





Database Trends: Findings & Implications

Findings

PostgreSQL is both the most used (11,514) and the most wanted (12,193) database

MySQL is second most used (8,556), but interest falls to 6,204

Redis rises from 5,814 to 6,384; Supabase enters the wanted list (1,623) while Oracle drops out

Implications

PostgreSQL is the strongest database skill to learn or adopt

MySQL is still widely used, but interest may be shifting toward PostgreSQL

Redis and Supabase are worth watching as growing options

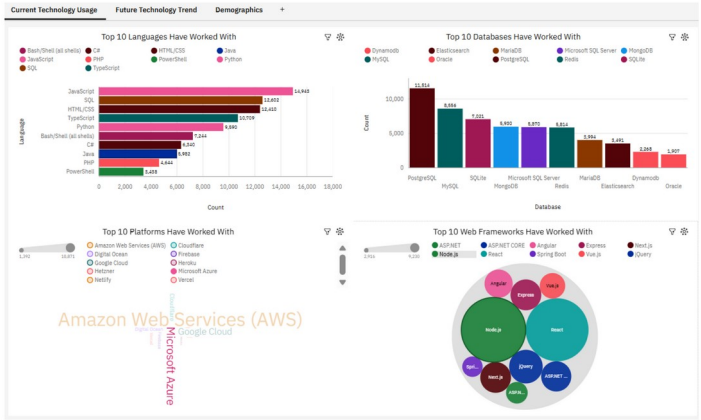
Dashboard

Interactive Cognos Analytics dashboard with three tabs

Tab 1: Current technology usage

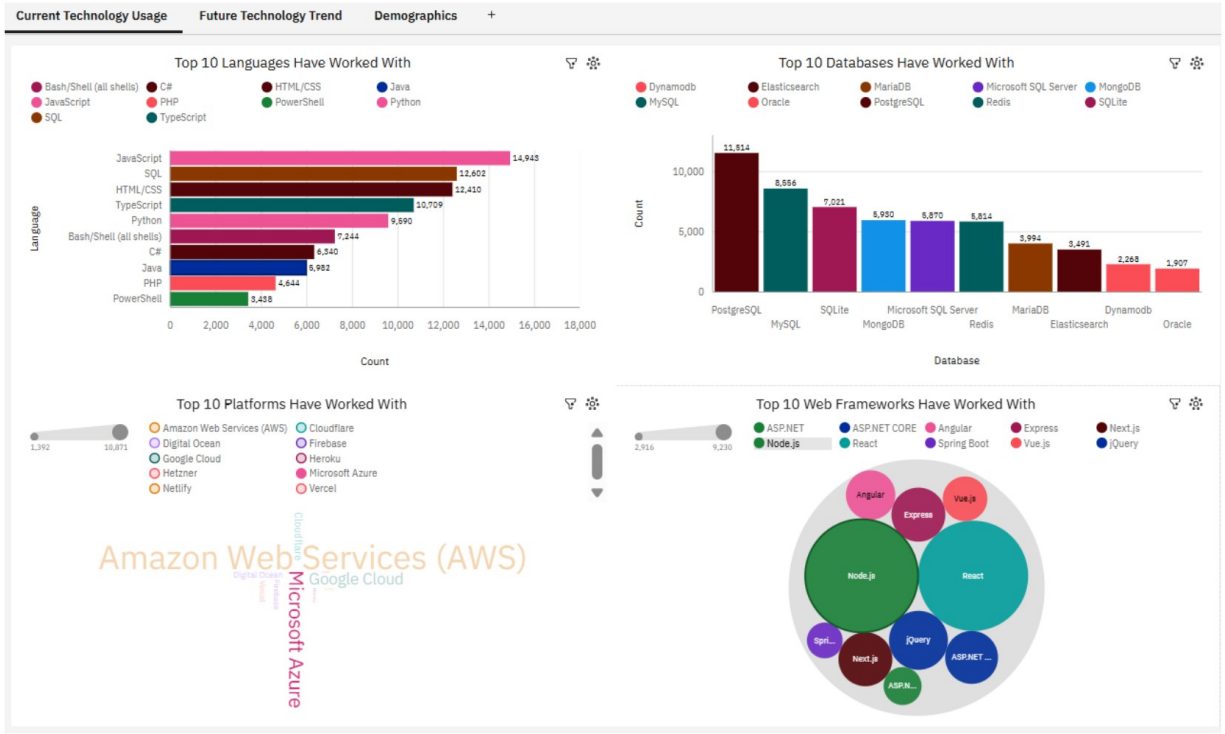
Tab 2: Future technology trend

Tab 3: Demographics

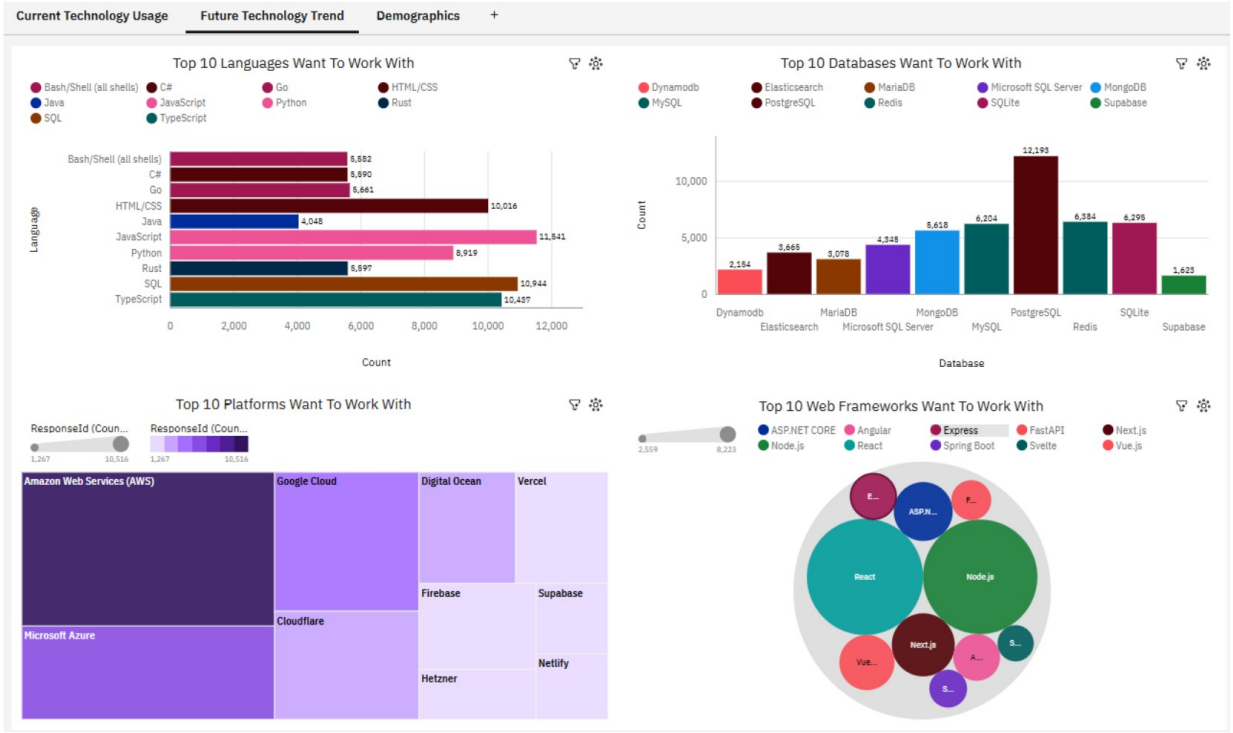


[View the survey dashboard (PDF)]

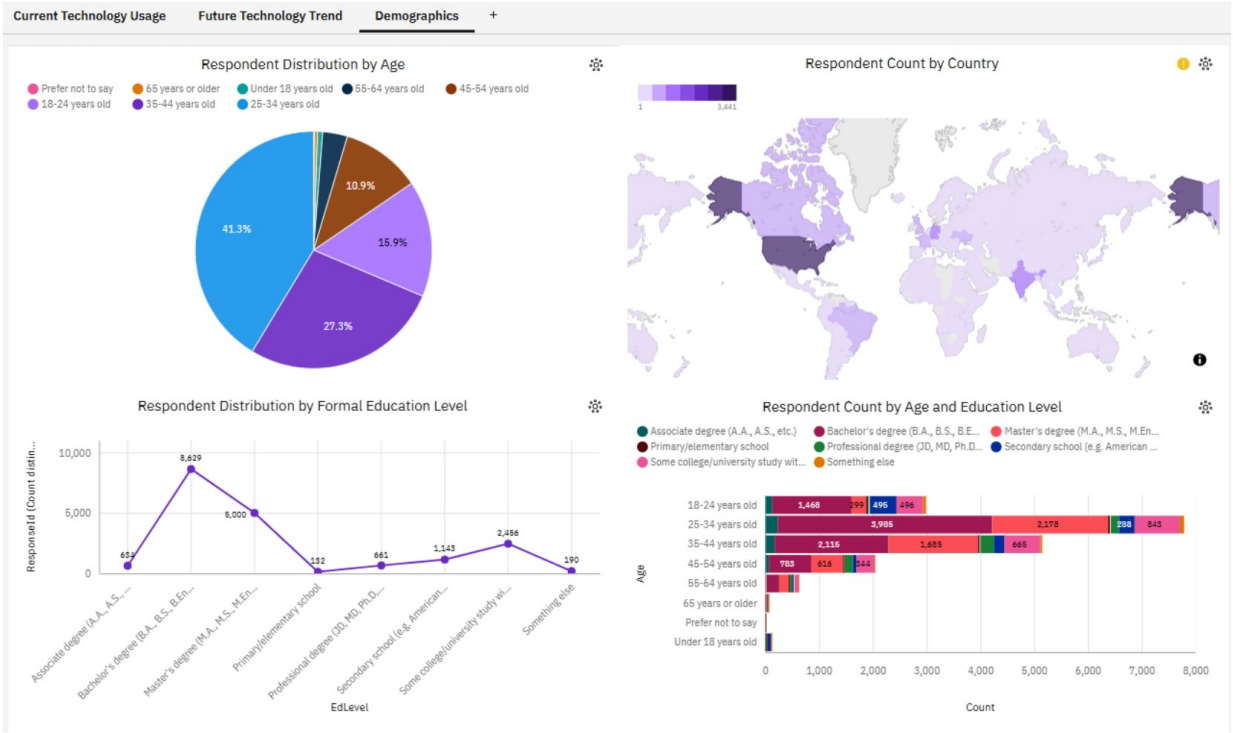
Dashboard Tab 1: Current Technology Usage



Dashboard Tab 2: Future Technology Trend



Dashboard Tab 3: Demographics



Discussion

1. **Stable core:** JavaScript and SQL lead both current use and future interest, with HTML/CSS and TypeScript close behind

2. **Database shift:** PostgreSQL is the most wanted database (12,193), while interest in MySQL falls from 8,556 to 6,204

3. **Emerging tools:** Rust, Go, Supabase and FastAPI show growing interest, though they are still smaller than the established tools

4. **Cloud:** AWS leads current (10,871) and future (10,516) platforms, with Azure and Google Cloud next

5. **Limitation:** 68.6% of respondents are aged 25-44 and most are professional developers, so findings may not describe all developers

Overall Findings & Implications

Findings

JavaScript and SQL lead both current use and future interest

PostgreSQL, AWS, Node.js and React lead their categories today and for the future

Rust, Go, Supabase and FastAPI are rising; 25-34 year olds are the largest group (41.3%)

Implications

Web and SQL skills are the safest foundation to build on

PostgreSQL, AWS and the Node.js/React stack are strong choices to learn or adopt

Emerging tools are worth watching, and findings mainly reflect younger developers

Conclusion

1. **Core skills:** JavaScript and SQL lead both current use and future interest
2. **Leading tools:** PostgreSQL, AWS, Node.js and React lead their categories today and for the future
3. **Emerging trends:** Rust, Go, Supabase and FastAPI show growing interest and are worth watching
4. **Next steps:** Track trends across multiple survey years and add job posting data to confirm employer demand

Appendix

Dashboard: github.com/ananpkumar/ibm-data-analyst-capstone (dashboard folder)

Data source: Stack Overflow Developer Survey. Notebooks use the full survey; the dashboard uses a course-provided extract of 18,845 responses

Tools used: SQL, Python (Jupyter Notebooks), IBM Cognos Analytics

Code and notebooks: github.com/ananpkumar/ibm-data-analyst-capstone